WIKIPEDIA
The Free Encyclopedia

Vachellia nilotica

Vachellia nilotica, more commonly known as ***Acacia nilotica***, and by the vernacular names of **gum arabic tree**,[5] **babul**,[6] **thorn mimosa**, **Egyptian acacia** or **thorny acacia**,[7] is a flowering tree in the family Fabaceae. It is native to Africa, the Middle East and the Indian subcontinent. It is also considered a 'weed of national significance' and an invasive species of concern in Australia, as well as a noxious weed by the federal government of the United States.[8]

Vachellia nilotica	
Scientific classification	
Kingdom:	Plantae
Clade:	Tracheophytes
Clade:	Angiosperms
Clade:	Eudicots
Clade:	Rosids
Order:	Fabales
Family:	Fabaceae
Subfamily:	Caesalpinioideae
Clade:	Mimosoid clade
Genus:	*Vachellia*
Species:	***V. nilotica***
Binomial name	
Vachellia nilotica (L.) P.J.H.Hurter & Mabb.[1]	
Subspecies	
Vachellia nilotica subsp. *adstringens* (Schumach. & Thonn.) Kyal. & Boatwr.	
Vachellia nilotica subsp. *cupressiformis* (J.L.Stewart) Ali & Faruqi	
Vachellia nilotica subsp. *hemispherica* Ali & Faruqi	

Taxonomy

This species of tree is the type species of the Linnaean genus *Acacia*, which derives its name from Greek ἀκακία, *akakía*, the name given by early Greek botanist-physician Pedanius Dioscorides (c. 40–90 CE) to this tree as a medicinal, in his book *Materia Medica*.[9] The genus *Acacia* was long known not to be taxonomically monophyletic, and despite being the type species of that genus, *A. nilotica* has since been moved to the genus *Vachellia*, with the genus name *Acacia* being reserved for Australian species; the principle of priority, which would normally prevent such a taxonomic change, was waived with a majority vote by the International Botanical Congress in 2005. The renaming of the traditional *Acacia* to *Vachellia* remains controversial, especially in Africa, where *V. nilotica* is an iconic species and is widely referred to as "the acacia".[10] For the new classification of this and other species historically classified under genus *Acacia*, see *Acacia*.

The genus name *Acacia* derives from ἀκακία, which may come from the ancient Greek word ἄκις, *ákis*, "thorn", for its characteristic thorns,[11] or may have been borrowed from a pre-Greek language.[12] The specific epithet *nilotica* was probably given by Linnaeus from the tree's originally known range along the Nile river. In Australia the tree is known as a **prickly acacia**,[13] despite usurping Dioscorides' two millennia-old etymology, the Australian species classified as *Acacia* in Australia do not have thorns.

Description

Spring blossoms at Hodal in Faridabad District of Haryana, India

Acacia nilotica or *Vachellia nilotica* is a tree 5–20 m high with a dense spheric crown, stems and branches usually dark to black coloured, fissured bark, grey-pinkish slash, exuding a reddish low quality gum. The tree has thin, straight, light, grey spines in axillary pairs, usually in 3 to 12 pairs, 5 to 7.5 cm (3 in) long in young trees, mature trees commonly without thorns. The leaves are bipinnate, with 3–6 pairs of pinnulae and 10–30 pairs of leaflets each, tomentose, rachis with a gland at the bottom of the last pair of pinnulae. Flowers in globulous heads 1.2–1.5 cm in diameter of a bright golden-yellow color, set up either axillary or whorly on peduncles 2–3 cm long located at the end of the branches. Pods are strongly constricted, hairy, white-grey, thick and softly tomentose. Its seeds number approximately 8000/kg.[14]

- *Vachellia nilotica* subsp. *indica* (Benth.) Kyal. & Boatwr.
- *Vachellia nilotica* subsp. *kraussiana* (Benth.) Kyal. & Boatwr.
- *Vachellia nilotica* subsp. *leiocarpa* (Brenan) Kyal. & Boatwr.
- *Vachellia nilotica* subsp. *nilotica* (L.) P.J.H.Hurter & Mabb.
- *Vachellia nilotica* subsp. *subalata* (Vatke) Kyal. & Boatwr.
- *Vachellia nilotica* subsp. *tomentosa* (Benth.) Kyal. & Boatwr.[2][3]

Range of *Vachellia nilotica*

Synonyms[4]

- *Acacia arabica* (Lam.) Willd.
- *Acacia nilotica* (L.) Willd. ex Delile
- *Acacia scorpioides* (L.) W.Wight
- *Mimosa arabica* Lam.
- *Mimosa nilotica* L.
- *Mimosa scorpioides* L.

Distribution

Acacia nilotica or *Vachellia nilotica* is native from Egypt, across the Maghreb and Sahel, south to Mozambique and KwaZulu-Natal, South Africa, and east through the Arabian Peninsula to the Indian subcontinent and Burma. It has become widely naturalised outside its native range including Zanzibar and Australia.[15] It is spread by livestock.[13]

Uses

Forage and fodder

In part of its range smallstock consume the pods and leaves,[16] but elsewhere it is also very popular with cattle. Pods are used as a supplement to poultry rations in India. Dried pods are particularly sought out by animals on rangelands. In India branches are commonly lopped for fodder. In West Africa, the pods and leaves are considered to have anthelminthic properties on small ruminants and this has been confirmed by in vitro experiments on nematodes.[16] In Kano of Nigeria, acacia pods have traditionally been used to dye leather a reddish-tinge.[17]

Tooth brushing

The tender twig of this plant is used as a toothbrush in south-east Africa and the Indian subcontinent.[18]

Seed pods

Gum arabic

The exudate gum of this tree is known as gum arabic and has been collected from the pharaonic times for the manufacture of medicines, dyes and paints. In the present commercial market, gum arabic is defined as the dried exudate from the trunks and branches of *Senegalia (Acacia) senegal* or *Vachellia (Acacia) seyal* in the family Leguminosae (Fabaceae).[19]:4 The gum of *A. nilotica* is also referred to in India as *Amaravati gum*.[20]

Gum arabic exuding

Lumber

The tree's wood is "very durable if water-seasoned" and its uses include tool handles and lumber for boats.[21] The wood has a density of about 833 kg/m^3.[2]

Food and medicine

In India it's used as an ingredient in various dishes.

The Maasai people eat both the inner bark (phloem) and the fruit pulp boiled in water. They also use this plant medicinally to treat sore throat, cough, chest pains etc.[22]

In Northern Nigeria it is called *bagaruwa* in Hausa. Medicinal uses include soaking the tender bark in water to be taken against dysentery and pile. The fruits are ground together with the seeds and taken with honey as treatment against stomach ulcers.

Trunk at Hodal in Faridabad District of Haryana, India

Phytochemistry

Two new antiprotozoal diterpenes have been isolated from the root bark of ***Acacia nilotica.*** [23]

Propagation

There are 5000–16000 seeds/kg.[24]

Gallery

Vachellia nilotica, Village Behlolpur, Punjab, India

village Chaparr Chirri, Mohali, Punjab, India

Compound leaf, seed, flower and seed pod

See also

- List of Indian timber trees
- Arid Forest Research Institute (AFRI)
- Babool (brand) of toothpaste
- Teeth cleaning twig(datun)

References

1. Kyalangalilwa B, Boatwright JS, Daru BH, Maurin O, van der Bank M (2013). "Phylogenetic position and revised classification of *Acacia s.l.* (Fabaceae: Mimosoideae) in Africa, including new combinations in *Vachellia* and *Senegalia*" (https://doi.org/10.1111%2Fboj.12047). *Bot J Linn Soc*. **172** (4): 500–523. doi:10.1111/boj.12047 (https://doi.org/10.1111%2Fboj.12047). hdl:10566/3454 (https://hdl.handle.net/10566%2F3454).
2. Wickens, G.E. (1995). "Table 2.1.2 The timber properties of *Acacia* species and their uses" (http://www.fao.org/docrep/V5360E/v5360e0f.htm). *Role of Acacia species in the rural economy of dry Africa and the Near East* (http://www.fao.org/docrep/V5360E/v5360e00.htm#Contents). FAO Conservation Guide. Vol. 27. Food and Agriculture Organization of the United Nations. ISBN 978-92-5-103651-8.
3. "*Acacia nilotica*" (https://npgsweb.ars-grin.gov/gringlobal/taxonomydetail.aspx?id=312499). *Germplasm Resources Information Network*. Agricultural Research Service, United States Department of Agriculture. Retrieved 12 December 2017.
4. "Acacia nilotica" (http://www.ildis.org/LegumeWeb?version~10.01&LegumeWeb&tno~323&genus~Acacia&species~nilotica). *LegumeWeb*. International Legume Database & Information Service.
5. "*Acacia nilotica*" (https://www.itis.gov/servlet/SingleRpt/SingleRpt?search_topic=TSN&search_value=182086). Integrated Taxonomic Information System.

6. "Definition of BABUL" (http://www.merriam-webster.com/dictionary/babul). *www.merriam-webster.com*. Retrieved 2017-08-03.
7. "*Vachellia nilotica*" (https://npgsweb.ars-grin.gov/gringlobal/taxonomydetail.aspx?id=465206). *Germplasm Resources Information Network*. Agricultural Research Service, United States Department of Agriculture. Retrieved 30 June 2017.
8. *Federal Noxious Weed List} web* (https://www.aphis.usda.gov/plant_health/plant_pest_info/weeds/downloads/weedlist.pdf) (PDF)
9. "*Acacia nilotica* (acacia)" (https://web.archive.org/web/20100112043913/http://www.kew.org/plants-fungi/Acacia-nilotica.htm). *Plants & Fungi*. Royal Botanic Gardens, Kew. Archived from the original (http://www.kew.org/plants-fungi/Acacia-nilotica.htm) on 12 January 2010. Retrieved 28 January 2010.
10. Kull, Christian A.; Rangan, Haripriya (2012). "Science, sentiment and territorial chauvinism in the acacia name change debate" (https://www.jstor.org/stable/j.ctt24h85b.11). In Haberle, Simon G.; David, Bruno (eds.). *Peopled Landscapes*. Archaeological and Biogeographic Approaches to Landscapes. Vol. 34. ANU Press. pp. 197–220. ISBN 978-1-921862-71-7. JSTOR j.ctt24h85b.11 (https://www.jstor.org/stable/j.ctt24h85b.11). Retrieved 2022-04-12.
11. Quattrocchi, Umberto (2000). *CRC World Dictionary of Plant Names* (https://books.google.com/books?id=esMPU5DHEGgC). Vol. 1 A–C. CRC Press. p. 6. ISBN 978-0-8493-2675-2.
12. Beekes, Robert (2010). "ἀκακία". *Etymological Dictionary of Greek*. Brill. p. 46.
13. "Prickly acacia – *Acacia nilotica*". Weed Management Guide (http://www.weeds.org.au/WoNS/pricklyacacia/docs/Weed_Management_Guide-Prickly_Acacia.pdf) (PDF) (Report). Weeds of National Significance. 2003. ISBN 978-1-920932-14-5 – via weeds.org.au.
14. "handbook on seeds of dry-zone acacias" (http://www.fao.org/docrep/006/Q2190E/Q2190E10.htm). *www.fao.org*. Retrieved 2017-08-03.
15. Kriticos, Darren; Brown, Joel; Radford, Ian; Nicholas, Mike (1999-10-01). "Plant Population Ecology and Biological Control: Acacia nilotica as a Case Study" (https://www.sciencedirect.com/science/article/pii/S1049964499907466). *Biological Control*. **16** (2): 230–239. Bibcode:1999BiolC..16..230K (https://ui.adsabs.harvard.edu/abs/1999BiolC..16..230K). doi:10.1006/bcon.1999.0746 (https://doi.org/10.1006%2Fbcon.1999.0746). ISSN 1049-9644 (https://www.worldcat.org/issn/1049-9644).
16. Zabré, Geneviève; Kaboré, Adama; Bayala, Balé; Katiki, Luciana M.; Costa-Júnior, Lívio Martins; Tamboura, Hamidou H.; Belem, Adrien M.G.; Abdalla, Adibe L.; Niderkorn, Vincent; Hoste, Hervé; Louvandini, Helder (2017). "Comparison of the in vitro anthelmintic effects of *Acacia nilotica* and *Acacia raddiana*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5703060). *Parasite*. **24**: 44. doi:10.1051/parasite/2017044 (https://doi.org/10.1051%2Fparasite%2F2017044). PMC 5703060 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5703060). PMID 29173278 (https://pubmed.ncbi.nlm.nih.gov/29173278).
17. Dalziel, J.M. (1926). "African Leather Dyes". *Bulletin of Miscellaneous Information*. **6** (6). Royal Botanic Gardens, Kew: 225–238. doi:10.2307/4118651 (https://doi.org/10.2307%2F4118651). JSTOR 4118651 (https://www.jstor.org/stable/4118651).
18. Saurabh Rajvaidhya *et al.* (2012) "A review on *Acacia Arabica*, an Indian medicinal plant" (http://www.ijpsr.com/V3I7/13%20Vol.%203,%20Issue%207,%20July%202012,%20RE-654,%20Paper%2013.pdf) *International Journal of Pharmaceutical Sciences and Research* Vol 3(7) pp 1995-2005
19. "Production and marketing of gum arabic" (http://ngara.org/Production%20and%20MArketing%20of%20Gum%20Arabic_english_french.pdf) (PDF). Nairobi, Kenya: Network for Natural Gums and Resins in Africa (NGARA). 2004.
20. "Acacia nilotica (gum arabic tree)" (http://www.cabi.org/isc/datasheet/2342). *Invasive species compendium*. Centre for Agriculture and Biosciences International. Retrieved 24 January 2016.

21. Mueller, Ferdinand (1884). "*Acacia longifolia*, Willdenow" (https://books.google.com/books?id=CfDcl2m-6BMC&q=%22acacia+longifolia%22+uses&pg=PA7). *Select extra-tropical plants readily eligible for industrial culture or naturalization*. G.S. Davis. p. 7.
22. Ruffo, Christopher K.; Birnie, Ann; Tengnäs, Bo (2002). *Edible wild plants of Tanzania*. Regional Land Management Unit/Sida. ISBN 9966-896-62-7.
23. Anyam, John V.; Daikwo, Priscilla E.; Ungogo, Marzuq A.; Nweze, Nwakaego E.; Igoli, Ngozichukwuka P.; Gray, Alexander I.; De Koning, Harry P.; Igoli, John O. (2021). "*Two New Antiprotozoal Diterpenes From the Roots of* **Acacia nilotica**"" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8097170). Frontiers in Chemistry. ***9**: 624741.* *Bibcode:2021FrCh....9...76A (https://ui.adsabs.harvard.edu/abs/2021FrCh....9...76A). doi:10.3389/fchem.2021.624741 (https://doi.org/10.3389%2Ffchem.2021.624741). PMC 8097170 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8097170). PMID 33968894 (https://pubmed.ncbi.nlm.nih.gov/33968894).*
24. "*Vachellia nilotica* (as *Acacia nilotica*)" (http://www.tropicalforages.info/key/Forages/Media/Html/Acacia_nilotica.htm). Tropical Forages.

External links

- Carter, J.O. (1998). "7.2 *Acacia nilotica*: a Tree Legume out of Control" (http://www.fao.org/ag/agp/agpc/doc/Publicat/Gutt-shel/x5556e0v.htm). In Gutteridge, Ross C.; Shelton, H. Max (eds.). *Forage Tree Legumes in Tropical Agriculture* (http://www.fao.org/ag/agp/agpc/doc/Publicat/Gutt-shel/x5556e00.htm#Contents). The Tropical Grassland Society of Australia. ISBN 978-0-9585677-1-8.
- "*Vachellia nilotica* (as *Acacia nilotica* subsp. *indica* (Benth.) Brenan)" (http://florabase.calm.wa.gov.au/browse/flora?f=163&level=s&id=20757&type=l&PHPSESSID=948be459b92215063fcb287af7ed3099). *FloraBase the West Australian Flora*.
- *Vachellia nilotica* (as *Acacia nilotica*) (www.frienvis.nic.in) (http://www.frienvis.nic.in/babul.htm)
- *Acacia nilotica* (http://www.westafricanplants.senckenberg.de/root/index.php?page_id=13&preview=true&searchTextMenue=Acacia+nilotica&search=Wikitemplate) in West African plants – A Photo Guide. (http://www.westafricanplants.senckenberg.de/)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Vachellia_nilotica&oldid=1225386299"